Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Use Disclosure Form

Attach to assignments where AI was used in any permitted capacity.

1. Course & Student Info

Field	Value
Course:	CS423 / CSC13003 – Software Testing
Assignment ID:	HW#01
Assignment Title:	QA/QC Jobs · 20 Defects · Test a Physical Product
AI Use Category (1–5):	Category 5
Date:	01/06/2026
Student name:	ĐINH HOÀNG NAM
Student ID:	23127430

2. Disclosure Questions

1. AI tool(s) used:

List every AI tool used for this assignment (e.g., AI Tool (e.g., ChatGPT, Claude, Gemini), ChatGPT, GitHub Copilot, Cursor, Gemini).

ChatGPT, Gemini

2. Stage(s) of the assignment where AI was used:

Tick all that apply: [X] brainstorming [ ] outlining [ ] drafting [] feedback [X] revision [ ] coding [X] data analysis [X] visual design [ ] other (specify).

3. Main prompts or tasks given to the AI:

Paste the 2–3 most impactful prompts verbatim. For the full transcript, attach Appendix A (prompt_log.md).

#Prompt 1

Hãy giải thích cho tôi về sự cố phần mềm [Tên sự cố] xảy ra vào năm [Năm xảy ra] liên quan đến [Tên công ty/Hệ thống]. Cung cấp nguyên nhân kỹ thuật, hậu quả và các đường link tham chiếu chính thức nếu có. Hãy điền cho tôi 20 thông tin vào đây để thành câu prompt để tôi có thể bẫy AI và đưa thông tin lại cho bạn

#Prompt 2:

Bug #[Số thứ tự]: [Tên sự cố phần mềm] ([Năm xảy ra])

Phân loại: [Ghi "Phần mềm truyền thống" HOẶC "Hệ thống AI/LLM" - Đảm bảo có >= 5 lỗi AI/LLM]

Đường dẫn nguồn gốc (Source Link): [Chèn link bài báo hoặc tài liệu kỹ thuật gốc mà bạn tự tìm được]

*Mô tả lỗi (Description):** [Tóm tắt ngắn gọn lỗi này là gì, xảy ra ở phân hệ nào của phần mềm]*

*Độ nghiêm trọng (Severity):** [Low / Medium / High / Critical - dựa trên thiệt hại hoặc chuẩn CVSS nếu có]*

*Hậu quả (Consequences):** [Hệ thống ngừng hoạt động bao lâu, thiệt hại tài chính, rò rỉ dữ liệu của bao nhiêu người dùng...]*

*Giải pháp khắc phục (Solution):** [Đội ngũ kỹ sư đã vá lỗi này như thế nào? Ví dụ: cập nhật phiên bản, sửa lại logic code, rollback hệ thống...]*

Điểm AI bị ảo tưởng/thiên vị (AI Mistake/Hallucination Instance):

Bằng chứng từ AI: [Trích dẫn lại chính xác câu văn hoặc đoạn thông tin sai lệch mà AI đã trả lời bạn trong quá trình tra cứu]

Phân tích phản biện : [Đưa ra phân tích dựa trên bằng chứng của AI]

Đây là format mẫu cho mỗi lỗi, hãy điền vào đầy đủ các thông tin liên quan đến lỗi và ghi nhận lại những thông tin Gemini cung cấp mà gây sai lệch hoặc thiên vị về từng lỗi

4. Specific parts of the work AI contributed to:

Be specific. Example: 'AI generated TC01–TC15 in Section 3.2; I rewrote TC04 and TC11; AI did NOT contribute to Sections 1, 2, 4, or the AI Critique.'

AI Contributed to:

- **Requirement 1 (AI Impact Analysis):** AI initially generated the draft text analyzing the impact of technology on 10 QA/QC job positions.
- **Requirement 2 (Defects Explanation):** AI (Gemini) generated the core technical descriptions, consequences, and resolution frameworks for the 20 historical software defects. After that, I completely cross-checked all 20 entries with Chat GPT against official vendor advisories, fixed technical hallucinations (such as correcting the LangChain function from eval() to exec(), fixing the LEMURLOOT webshell name, and replacing fake URLs with authentic post-mortem source links), and

authored the "AI Mistake/Hallucination Instance" and "Phân tích phản biện" sections entirely by myself.

- **Requirement 3 (Test Cases Design):** AI generated the baseline structure, steps, and expected results for the functional test cases (TC01–TC15) in the keyboard testing suite.
- **Appendix A (Mindmap Blueprint):** AI generated the initial hierarchy and raw text format for the ISTQB process mindmap.

Student Modifications & Human-written parts:

- **Requirement 1:** I reviewed the job descriptions, identified the digital management tools, and optimized the text layouts by bolding specific technical constraints.
- **Requirement 2:** I completely cross-checked all 20 entries against official vendor advisories, fixed technical hallucinations (such as correcting the LangChain function from eval() to exec(), fixing the LEMURLOOT webshell name, and replacing fake URLs with authentic post-mortem source links), and authored the **"AI Mistake/Hallucination Instance"** and **"Phân tích phản biện"** sections entirely by myself.
- **Requirement 3:** I structurally modified the entire test suite by adding two mandatory operational columns (*Actual Result* and *Verdict*) to ensure alignment with the FIT-HCMUS STLC execution guidelines, and I completely formulated and highlighted 3 complex physical hardware edge cases (TC16–TC18).
- **Appendix A (Mindmap Rectification):** I thoroughly audited the raw mindmap structure, discarded AI's hallucinated phases, and reconstructed a correct 6-phase STLC Mermaid diagram matching the **Slide S02** core specification.
- **AI-free Categories:** AI did **NOT** contribute to the physical device photo verification, the 5 demonstration video logs, the compilation of the job advertisements screenshots, the file prompt_log.md, or the final **AI Critique (Section 5: Conclusion)**.

5. How I reviewed, revised, or verified the AI output:

Describe your verification method (ran the test, checked the spec, asked the TA, looked up RFC, cross-checked with the ISTQB syllabus, etc.)

- I applied a cross-AI verification method combined with independent auditing using original documents. After ChatGPT generated the mind map code, I sent the entire code to the Gemini model with a command requesting a knowledge error check. Gemini issued three warnings about incorrect phase classification. To verify, I directly opened the lecture material Slide S02_Software Testing Life Cycle from the IT Department of HCMUS. By directly comparing it with the phase diagram on page 5 and the deliverables list on pages 7 and 11, I confirmed that Gemini's critique was completely accurate and ChatGPT's output contained incorrect academic knowledge. From there, I manually corrected the entire diagram using Mermaid code.

- My validation method uses cross-AI verification combined with real-world physics experiments. First, I took the 15 test cases generated by ChatGPT and fed them into the Gemini model for error checking. After Gemini pointed out weaknesses in edge coverage, I independently compared them with Slide S02 from the FIT@HCMUS course to confirm the test table structure lacked the Actual Result and Verdict columns. Finally, I directly executed these scenarios on my mechanical keyboard to fill in the actual results and recorded 5 short narration videos under.

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6. Citation (if required by course style guide):

Software Testing uses the IEEE style. Example: Anthropic. (2026). AI Tool (e.g., ChatGPT, Claude, Gemini) [Large language model]. https://claude.ai

[1] OpenAI. (2026). ChatGPT (Model GPT-5.5) [Large language model]. https://chatgpt.com

[2] Google. (2026). Gemini (Model Gemini 3.5 Flash) [Large language model]. https://gemini.google.com

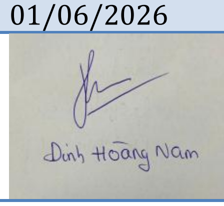
[3] Faculty of Information Technology - HCMUS. (2026). Slide Bài giảng Môn học Kiểm thử Phần mềm [Course Slides]. Ho Chi Minh City University of Science.

[4] International Software Testing Qualifications Board. (2023). Certified Tester Foundation Level Syllabus [Official Syllabus]. https://www.istqb.org

3. Statement of Honesty

By signing below, I confirm that the disclosure above is accurate and complete. I understand that undisclosed or false disclosure of AI use is treated as academic misconduct and may result in a 0 grade for the assignment and disciplinary referral.

Signature

Student name (printed):	**ĐINH HOÀNG NAM**
Student ID:	23127430
Class / Cohort:	23KTPM1
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Dr. Lam Quang Vu / MSc. Truong Phuoc Loc / MSc. Ho Tuan Thanh
Date:	01/06/2026
Signature:	

References

- Kharbach, M. (2026). AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0.
- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338.
- Perkins, M., Roe, J., & Furze, L. (2025). AI Assessment Scale.
- Anthropic (2025). Building reliable AI test agents — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.